

Game Makers — Week 4

Decisions and Sensing

Name: _____

TODAY'S BIG IDEA

The computer can ask a **yes-or-no question** about the world, and do different things depending on the answer.

```
if <touching the player?> then
  ...do this...
else
  ...do that instead...
```

A question that can only be answered yes or no is called a **boolean**. Those blocks are **pointy hexagons**, and they only fit in pointy hexagon slots. Shape tells you what fits where.

WORDS TO KNOW

Word	What it means
condition	The yes/no question
boolean	An answer that can only be yes or no (true or false)
if / then	Do something only if the answer is yes
if / then / else	Do one thing if yes, a different thing if no
sensing	Blocks that let a sprite notice the world

 **Big one today:** an **if** block checks **once**, at the moment it runs. If you want it to keep checking, it has to be **inside a loop**.

CLASS ACTIVITY A — TRAFFIC LIGHT

```
when green flag clicked
  forever
    if <touching [Red]?> then
      say [STOP!]
    else
      say [GO!]
```

Now drag your sprite in and out of the red one.

CLASS ACTIVITY B — BUMPER SPRITE

```
when green flag clicked
  forever
    move 5 steps
    if <touching [edge]?> then
      turn 180 degrees
    wait 0.01 seconds
```

Add a second sprite and make them bounce off each other too.

YOUR PROJECT — CATCHING! 🎉

Today your project becomes a real game you can actually play.

The new bit goes **inside the falling loop**:

```
repeat 20
  change y by -18
  wait 0.02 seconds
  if <touching [Player]?> then
    start sound [Pop]
    go to x: (pick random -220 to 220) y: 170
```

- [] Catching works
- [] A sound plays when you catch something
- [] Playtest it — then swap seats and play your neighbour's!
- [] **Save it**

Homework

1. Catch feedback. Make catching do something noticeable — a flash, a costume change, a different sound, a `say`.

2. Something you shouldn't catch. Add a bad falling object (a bomb, a coconut, a cat). For now it just says "Ouch!" when it hits you. Next week it'll cost you points.

Bonus challenges

1. A sprite that says different things depending on which half of the stage it's on. (Hint: `if <(x position) > 0>`)
2. A sprite that follows the mouse, but only while space is held down.
3. Use `if` to stop your player leaving the stage — the thing we couldn't do in week 2!
4. Use `and` to combine two conditions: only catch it if you're touching it **and** pressing space.

IF SOMETHING BREAKS

- **Catching never works?** The `if` isn't inside the loop.
- **Works once, then never again?** The `if` is in the outer `forever` but not in the `repeat`.
- **Object falls straight through you?** You need the `go to` inside the `if` to send it back up.
- **Catching is really fussy?** Your object moves too far between checks. Make `change y by` smaller and the `repeat` number bigger.